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Glass welding with ultra-short laser pulses and long focal lengths – A process model

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Abstract

The welding of glasses with ultra-short laser pulses and high-NA optics has matured to a well-established process. The short focal length requires relatively thin work pieces in the order of 1-2 mm, and the short Rayleigh length requires a very precise adjustment of the laser focus relative to the interface to be welded.

To make welding of thick samples possible with relaxed alignment requirements, we welded optically contacted fused silica with a focal length of 65 mm. The resulting welding seams can be as deep as 2 mm and more. It turned out, that they grow from the focal point towards the laser source with increasing pulse energy and irradiation time. A thermal model for the growth of the molten zones will be presented and compared to the process with short focal lengths. Furthermore, the results of tensile tests and helium-leakage tests of welded fused silica samples will be presented.

Keywords: glass; welding; USP laser; process model

1. Introduction

Glass is a unique material considering its chemical resistance, dielectric strength, optical transparency and the possibility of optical imaging. These technical properties make it indispensable for the application of modern technology. However, the joining of glasses is still a challenge. Especially if the joining zone has to have the same properties as the base material, the classic joining techniques like gluing, soldering and heat-diffusion welding cannot be used in many applications.

In recent years, micro glass welding with ultra-short pulse lasers (USP) has been discussed in scientific publications [1, 2, 3]. This process is based on a spatially limited energy input, which creates molten zones with a diameter of approximately 100 µm. By lining up these welding zones, flat connections can be established. Since the process works through the transparent glass, interfaces can be welded that are not directly accessible. Due to the very small heat affected zone, glasses with very different expansion coefficients and assemblies with temperature-sensitive elements can also be joined. One of the biggest limitations of this joining technology is that the interfaces of the joining partners have to be in optical contact beforehand, which places high demands on the surface quality of the joining zone. As shown later by several groups, the USP-welding process is able to bridge and even reduce gaps of several micrometers between the interfaces [2, 4].

Another hurdle to the industrial usage of the process is its productivity. Fixed, high-numerical-aperture (high-NA) optics are commonly used, and the glasses are moved with linear axes, which limits the usable laser powers, possible feed rates and work piece thickness. The high-NA optics and low laser pulse energies of up to several micro joules are used to enable optical breakdown in the glass while circumventing self-focusing [5].

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The classical UPS-welding process described in the literature is based on optical breakdown. Therefore, the laser radiation is focused into the transparent material. At optical intensities in the order of 10^{13} W/cm^2 (depending on wavelength and material) multiphoton absorption occurs and generates free electrons. These electrons can then directly absorb more energy from the light field. On a timescale of several picoseconds, the electrons transfer their heat to the glass lattice. At high repetition rates, subsequent pulses can heat the glass lattice to the melting point while the low thermal conductivity limits the molten zone well below millimeter-size. For the initial focusing, high-NA objectives are used to prevent the ablation of the upper glass surface by lowering the optical intensity there through the high convergence. In addition, the high NA allows to reach the intensity threshold for optical breakdown while staying below the power threshold for self-focusing.

Publications on the welding under self-focusing conditions on the other hand are rare: As described in [6], it has been used for reinforcing direct bonds, increasing their shear strength by a factor of three. In [7], the authors describe experiments on long-focal USP glass welding. They report on welding speeds of up to 1 m/s in fused silica and strengths of up to 30 MPa, but without details on the strength-measurement or connections to the laser parameters.

In previous publications [8,9] we demonstrated that the process of welding with a long focal length of 65 mm starts when the pulse energy respectively the peak power exceeds the critical power for self-focusing. Above that energy the length of the modification increases linearly with the pulse energy as long as the feed rate is low enough for the heat accumulation to melt the whole length of the filament. Molten zones of 2 mm length were generated. Furthermore, the position of these molten zones is stable and controllable despite the fact that the process is dominated by Kerr lensing. Finally, it was shown that the bonding strength of the weld changes continuously when different sections of the filaments are used for welding the interface. This greatly relaxes the positioning requirements of the process. 20% of the tensile strength of the fused silica base material were reached with an un-optimized weld geometry. The higher process speeds enabled by galvoscanners are intended to increase the economics of the process.

The goal of the following work was the further optimization of the welding with long focal lengths and a deeper understanding of the generation of the very long welds.

2. Setup

The experiments described here were conducted with an “Amphos 100 flex” laser system with pulse energies up to 10 µJ, a repetition rate of 1 MHz, a wavelength of 515 nm and a pulse duration of 700 fs. The laser beam was directed through a scanlab intelliscan 14 galvoscanner. An F-theta lens of 65 mm focal length focused the beam down to a spot size of 12 µm including the enlargement by the refractive index of the work piece but not including the effect of the Kerr lens.

UV-grade fused silica was chosen as the work piece material. Mirror substrates of 1-inch diameter and 6 mm thickness with a surface flatness of λ/8 and 20/40 S-D were used. They were cleaned with isopropanol, blown dry, placed and gently pressured on each other by hand for optical contacting. No heat was applied. For further details of the sample processing see [8, 9].

After the welding, the samples were examined in different ways: Some of the samples were cut for optical microscopy of the welding seams. For this purpose, the edges were submerged in glycerol for index matching in order to avoid polishing. The rest of the samples was glued with DELO AD840 adhesive to M6 stainless steel DIN 464 knurled nuts for testing on a tensile testing machine (Instron 4411). Both knurled nuts were connected with M6 threaded bars with a length of 120 mm, and roughly the last 40 mm of the bars were clamped into the tensile testing machine. With this measure, it was ensured that the flexibility of the bars would largely compensate for any shear forces caused by a lack of centricity of the glued items.

3. Results

The central question is how the long welding zones are generated. The Rayleigh-length of our laser focus is in the order of 200 µm, much shorter than the molten zones. Previous experiments [8, 9] have also shown, that the welding process is dependent on self-focusing. Independent of the average power, the process will not start when the pulse energy is below the self-focusing threshold. Due to this self-focusing, filamentation will occur, which is the interplay between self-focusing and diffraction. That means that the beam is influenced by self-focusing, reducing the size of the initial focus. The high intensity in the focus will cause nonlinear absorption and plasma generation. The plasma acts as a diverging lens which increases the beam size. While leaving the plasma, the beam will self-focus again, defocus again, etc., until the energy loss by the plasma generation reduces the pulse energy either below the self-focusing threshold or below the threshold for substantial nonlinear absorption.

As a side note, it is very difficult the measure the exact position of the laser focus in the material during the experiment. This is because of the focus shift due to the refractive index of the material and due to the Kerr-effect (self-focusing). Because of that, it is very hard to tell, where the laser focus was relative to the molten zone visible under the microscope.

We compared our results with models in the literature, especially with the results of Nguyen [10]. Since the molten zones are thicker in the upper part, towards the laser beam, and the first plasma bubble in the plasma chain of the filament should absorb the most energy, our first process model was the following (see Fig.1):

Each consecutive laser pulse gets focused on the same spot by the f-theta lens and the Kerr-effect. The laser light of each pulse follows essentially the same optical path, which is formed by the mentioned interplay between self-focusing and plasma-diffraction. Upon the consecutive pulses, this optical path heats up and forms the molten zone, which grows in the travel-direction of the light (black arrow in Fig. 1), since the upper parts heat up stronger due to the higher pulse energy in the beginning of the filament. For higher feed-rates or lower laser powers, we observed shorter molten zones and roughly the

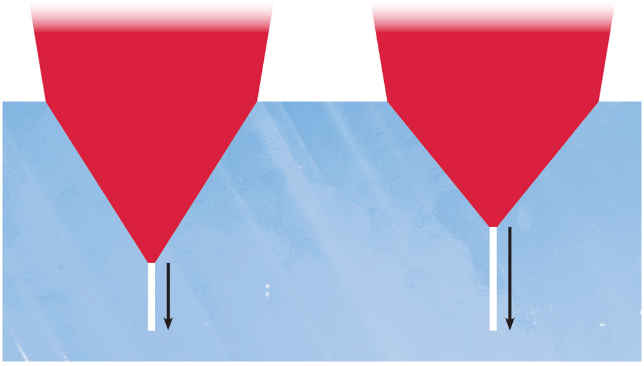

Fig.1. Preliminary process model, left: low pulse energies, right: high pulse energies.

same bottom positions as with higher pulse energies see [8, 9]. Our explanation was that in this case the optical path was only party molten since the deposited energy did not suffice to melt the whole filament-volume. The left part of Fig. 1 shows our preliminary process model for low pulse energies: Moderate Kerr-shift and short filament/molten zone. The right part shows the case of higher pulse energies: The Kerr-shift is stronger and the molten zone gets longer, since the higher pulse energy can generate a longer plasma channel. In this model, the thermal effects on the optical path are considered to be small compared to the Kerr-effect and the plasma-diffraction.

Using this process model, experiments were conducted to optimize the welding process, especially the speed. Therefore, the pulse energy and the feed-rate of the process were increased, with unexpected results, as shown in Fig. 2. Here, the laser focus was moved through the sample from left to right. The resulting molten zones are bent in a regular fashion. This experimental result does not fit with the preliminary process model. Laser filamentation is usually straight. Inhomogeneities in the material could disturb the filaments, but the regular bending seen here must be caused by the process itself. Judging by the feed direction, the molten zones could have grown upwards. Interestingly, the bending radius together with the continuous feed-rate shows, that the growth speed increases towards the top of the molten zone.

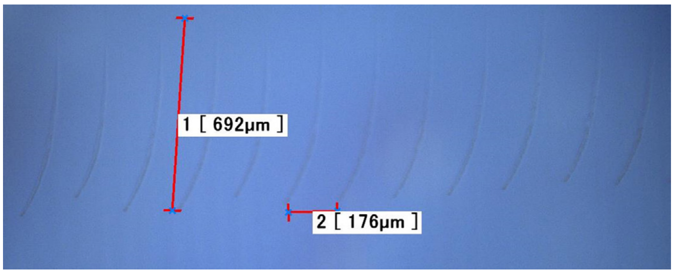

Fig. 2. Microscopic side view of a weld seam with high feed-rate.

To better understand this process, more welding lines were written, but this time the pulse energy was held constant at 5µJ. This way we are sure, that the focus-conditions including the Kerr-effect are the same for all welding lines. The feed-rate was set to 1mm/s and all other experimental conditions were held constant except for the repetition rate. The results can be seen in Fig. 3. As a consequence of the repetition rate being the only

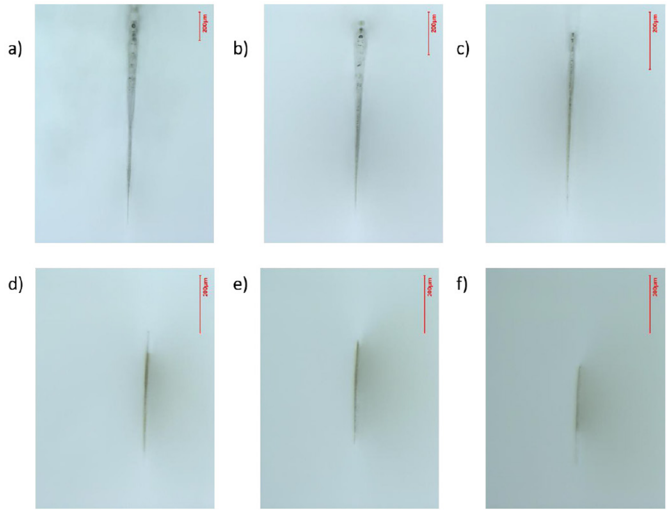

Fig. 3. Microscopic side view of welding lines with repetition rates of a) 2 MHz, b) 1 MHz, c) 800 kHz, d) 600 kHz, e) 400 kHz and f) 200 kHz, note the different length scales: the red bar is 200 µm in each of the pictures.

varied parameter, we can be sure that only the energy input and the thermal conditions can have changed. We can see, that the bottom of the molten zone is in the same height in all welding lines. The higher the repetition rate, the longer the molten zone is. Since the pulse energy is equal in all experiments this effect cannot be caused by the Kerr-effect as explained in the preliminary process model.

From these experimental data we conclude that the molten zone grows towards the laser source from the original laser focus. While the original focus position is affected by self-focusing, this effect would stay the same for consecutive pulses. To explain the growth of the molten zone, thermal effects have to be included, see Fig. 4. We propose that the first laser pulse hits the original focus position and causes a small plasma volume in the order of 10 µm (Fig. 4 left). Until the next laser pulse arrives, the plasma cools down and the heat diffuses into the vicinity of the focus. In addition to the constant Kerr-lens, the next laser pulse is affected by the thermal profile around the last laser focus. The lensing effect of this thermal distribution leads to stronger focusing and the focus shifts towards the laser source. Also, the higher temperature can generate free electrons, which would increase the laser absorption. Since the thermal distribution spreads out spherically, the molten zone can grow sideward, when the

Fig. 4. Final process model: on the left side the first laser pulse is absorbed and causes an initial thermal lens. On the right side consecutive pulses are focussed by this expanding thermal lens, so the focus is shifting towards the laser source.

geometrical laser focus moves, as seen in Fig. 2. The observation that the molten zone seems to grow faster in the upper part is explained by the fact, that the upper part is affected by the accumulated heat of many consecutive pulses. So the average temperature is higher, and following laser pulses can more easily melt bigger volumes. In Fig.2, when the laser beam travelled so far sideward that the thermal lens does not focus it, the process starts again at the original focal position.

With this process model, optical filamentation of the laser beam, like described in [10], is not necessary to explain the long molten zones. Probably the absorption in the laser plasma is strong enough that there is no pulse energy left after the first plasma, at least not enough energy to cause another Kerr-focus.

3.1. Optimization of the welding process

In the ongoing effort, to improve the welding quality, we tested a point-and-shoot process strategy instead of welding lines. With the galvoscanner used here, this is easily programmable and faster in the application. As already shown in the case of high-NA welding [11], the welding of single spots hinders the spreading of fractures, since they cannot follow the tension lines along the weld seams. Since welding surfaces are in optical contact before the welding, this process strategy does not change the leak tightness compared to line welding. The welding spot distance in a rectangular point-welding-pattern was varied and found to be optimal at around 100 µm, see inset in Fig. 5. We were able to weld the full area of 1-inch diameter fused silica substrates together in about 16 minutes. This resulted in a tensile strength of 28% of the base material. The details on the testing setup are described in [8, 9]. As Fig. 6 shows, this strength is very reliable over many samples. While the maximum demonstrated strength with this long focal length welding is lower than the values demonstrated for high-NA welding, the absolute value is sufficient even for demanding applications and about 30 times higher than direct bonding alone. The application of the rectangular pattern is very easy to parallelize, leaving plenty of room for higher process speeds.

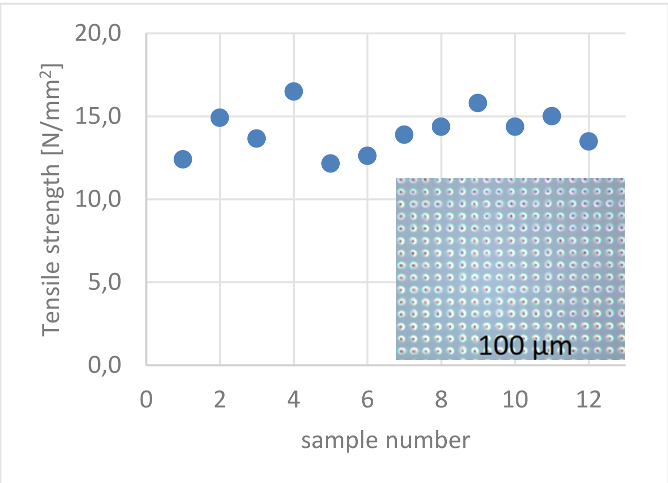

Fig. 5. Tensile strength of multiple samples of 1" diameter welded under the same conditions.

4. Conclusion

USP-welding with long a long focal length was investigated and process models were proposed. Our process model can explain the bending and the different growth speeds observed in welding experiments. Furthermore, the welding process itself could be optimized to show a tensile strength of 28% of bulk fused silica, with very little specimen spread. The long focal length process used here enables the use of galvoscanners to weld heavy pieces, that are hard to move with high precision axes. Also, the process enables the welding of thick work pieces: It was possible to work through a top layer of 26 mm thick fused silica.

Future work will concentrate on the quantification of the model. While the magnitude of thermal lensing is in the right order to explain the effects, specific calculations will be done to further support the model. Also, further experiments will be conducted to transfer the welding process to other materials.

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